

Resource Provisioning in a Cloud-Edge Computing Environment

This article demystifies the labyrinth of resource provisioning in a cloud-edge computing environment by looking at the concept, use cases and practical approaches.

Provisioning a resource in any type of a cloud environment is undoubtedly a daunting task.

Ensuring a fair share of resources while putting up a new workload on a cloud is complex but well understood as of now. However, the problem becomes challenging when we consider it in the context of cloud Internet of Things (IoT), where the resources need to be provisioned across the edge and the

cloud. The presence of two different systems with varying characteristics makes it an interesting subject to study.

In this article we look at the issues and the corresponding approaches, along with the concept of cloud-edge computing and its use cases.

The story so far

Cloud computing resource provisioning encompasses putting up various types of resources

like compute, storage, memory, network, and applications/ services/ microservices/ lambdas over the Internet. The proliferation of mobile technology makes things interesting. Nowadays our phones are equipped with good quality sensors, and hence all computations and their outcomes can potentially be context sensitive – both spatial and temporal. This results in a new variant, often loosely termed as mobile cloud computing.

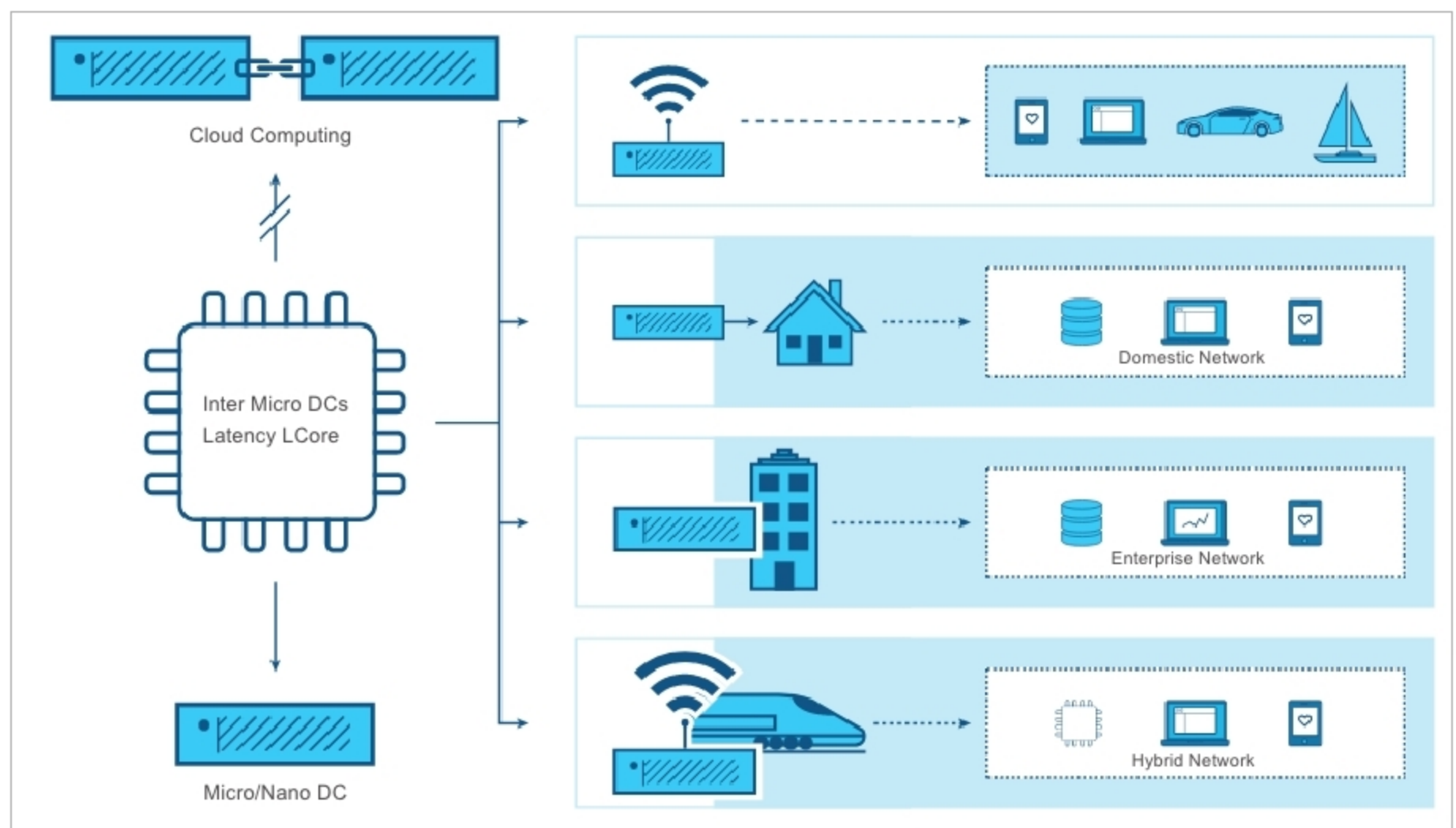


Figure 1: Cloud-edge computing model [Reference: <https://www.openstack.org/use-cases/edge-computing/cloud-edge-computing-beyond-the-data-center/>]

